

Advanced Approaches to Controlling Confounding

A plain-language guide to the study designs and analytic tools researchers use to make fair comparisons between patients on different drugs.

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Based on the chapter by Sebastian Schneeweiss and Samy Suissa, in *Pharmacoepidemiology*, Sixth Edition (Wiley)

Sicker Patients Often Get Different Drugs

Pharmacoepidemiology studies lean heavily on secondary data — insurance claims, medical records, disease registries — collected for billing or care, not research. The drug a patient received is very often tied to how sick they already were.

That link between health status and treatment is called confounding by indication. Left uncorrected, it can make a drug look more dangerous, or safer, than it really is. This chapter is about the tools researchers use to estimate a drug's true effect as validly as possible.

FOUNDATIONS, 2 OF 4

The Impossible Experiment Behind Every Design

Treat a patient, rewind time, leave them untreated, compare. That ideal comparison is impossible — every real design stands in for it, and the best stand-in depends on where the natural variation in who gets treated comes from.

Within One Patient

Sporadic use, a short washout, a fast side effect → case-crossover or crossover trial.

Between Patients

Some get drug A, others drug B, for reasons the data can't fully explain → cohort study or randomized trial.

Between Providers

Some hospitals or regions just favor one drug → instrumental variable analysis or cluster-randomized trial.

Two Strategies This Chapter Covers

1. Collect Better Data

Efficient sampling designs that gather rich confounder detail on a chosen subset of a large cohort, not the whole population.

2. Analyze More Cleverly

Analytic strategies — propensity scores, restriction, instrumental variables, sensitivity analysis — that get more validity out of data already collected.

We start with the most basic design choice: how a study cohort is built, and why the moment a patient starts a new drug matters so much.

THE NEW-USER DESIGN, 1 OF 4

Start the Clock When Treatment Starts

The new-user, or incident-user, design builds a cohort the way a randomized trial builds its patient group — minus the randomization. Every patient enters at the moment they start a new drug, compared against others also just starting a different one, all judged by a physician to need treatment at that same moment.

This section covers how new-user cohorts are built, two classic traps the design avoids, why chronic-disease studies often can't find enough drug-naïve patients, and how the “prevalent new-user” design fixes that with matched switchers.

THE NEW-USER DESIGN, 2 OF 4

Avoiding Two Classic Biases

Overadjustment

Measuring confounders only after treatment starts risks adjusting for factors that are themselves a result of the drug — biasing away part of the effect a study is trying to measure. Starting at first use avoids this.

Immortal Time Bias

A flawed shortcut labels patients “nonusers” if they skip a drug during some window. By definition they couldn’t have died or dropped out in that window — stacking the deck in favor of whichever group gets the head start.

A related reason to distrust comparisons against long-term users of an older drug: those patients are treatment survivors who already tolerated it, while patients susceptible to its early harms may have already stopped — so prevalent users can make the older drug look artificially safer.

THE NEW-USER DESIGN, 3 OF 4

When Truly New Users Are Hard to Find

Restricting to new users shrinks the eligible population. For second-line treatments of a chronic disease, that shrinkage can be severe: most patients arrive there only after already being treated with something else, so almost nobody is a genuinely drug-naïve new user.

Switching is rarely random — it usually follows disease progression, treatment failure, or side effects. So the fairest comparison is not switchers versus long-time first-line patients, but switchers to drug C versus switchers to drug D, matched at a similar point in their treatment history.

THE NEW-USER DESIGN, 4 OF 4

Matching Switchers to Comparable Initiators

Schneeweiss, Toh, Glynn, and colleagues proposed a fix: build a base cohort of everyone using either drug, including first-time starters and switchers alike. Each switcher is matched to a comparator user at the same point in their own treatment history — the same duration, the same number of prior prescriptions — an “exposure set.”

The matching uses a time-conditional propensity score, estimating switching-likelihood at each exposure-set time point, which keeps the matching intact and avoids adjusting away factors caused by treatment. Where sets are too large, conditional logistic regression on a random sample works instead. Bias from prevalent users can be checked by stratifying results by incident versus prevalent status.

SAMPLING DESIGNS, 1 OF 10

Why Sample Instead of Using Everyone?

Collecting detailed confounder data — chart reviews, questionnaires, lab values — on an entire large cohort is often too costly and slow. A study of antihypertensives and cancer risk followed 1.1 million patients for 14 years; 41,059 developed cancer. Reviewing full records for all 1.1 million was not feasible.

These designs pay off most when confounder data needs expensive chart review, interviews, biomarkers, or heavy processing. If everything already sits in the electronic database, analyzing the whole cohort costs little more than a sample.

This section covers three designs — nested case-control, multi-time case-control, and case-cohort — that collect rich data on a sample of a cohort without sacrificing validity.

SAMPLING DESIGNS, 2 OF 10

Two Ways to Slice a Cohort's Timeline

Every sampling design starts by defining a “risk-set” for each case: everyone still in the cohort and free of the outcome at that case’s index date. A cohort’s timeline can be organized two ways, changing who lands in each risk-set.

Calendar Time

Patients ranked by the actual date they entered the cohort. A risk-set is everyone who is under follow-up and outcome-free at the case’s calendar date — even patients who entered long before.

Follow-Up Time

Patients ranked by time since entry or start of exposure; a risk-set is everyone else at that same duration.

The same cases can end up with different risk-sets depending on the axis chosen — with full cohort data, researchers can switch the primary time axis to fit the question.

SAMPLING DESIGNS, 3 OF 10

The Nested Case-Control Design

Introduced by Nathan Mantel and developed into today's form by F.D.K. Liddell's asbestos and lung-cancer study, this design samples controls from within a cohort in four steps.

Step 1

Define the time axis

Step 2

Select all cases

Step 3

Form each case's risk-set

Step 4

Draw controls at random

A future case can still serve as a control for an earlier case, and the same subject can be drawn more than once — that's intentional, valid risk-set sampling, not an error. Repeated selection is intentional; restricting controls to permanent noncases is what creates the bias, by preferentially selecting people who happen to survive event-free for longer.

SAMPLING DESIGNS, 4 OF 10

Real Example: The Saskatchewan Asthma Cohort

A cohort of 12,301 asthma patients, followed from 1978 to 1987, produced 129 cases of death or near-death from asthma studied for their link to inhaled beta-agonists. A standard six-controls-per-case sample would have forced physicians to answer the same chart questions about the same patient across different windows.

The fix: finely stratify by age, area of residence, social assistance, prior hospitalization, and calendar date of entry, creating 129 subcohorts — one per case — each with two to eight controls. Every subject appeared only once, and the matched structure was preserved with conditional logistic regression.

SAMPLING DESIGNS, 5 OF 10

How Many Controls Do You Need Per Case?

One control per case is simplest but needs the most cases for power. Since case count is usually fixed, adding controls is often the only lever left. Example: 30% exposure prevalence in controls, detecting a true relative risk of 2, at 5% significance and 80% power.

Control-to-Case Ratio	Cases Needed	Total Size
1:1	122	244
2:1	90	270
4:1	74	370
10:1	65	715
20:1	62	1302

Gains are large up to about four controls per case, then flatten fast. Exceptions exist when exposure is rare: two studies in 40,000+ rheumatoid arthritis patients used 100 controls per case for rare drug-related hepatic and lung events.

SAMPLING DESIGNS, 6 OF 10

Getting More From the Same People

Instead of recruiting more people, the multi-time case-control design boosts precision by using more exposure time-windows per subject. The International Agranulocytosis and Aplastic Anemia Study collected four weeks of exposure history for its 221 cases and 1,425 controls, but early analyses used only one of those weeks.

Using all four weeks instead of one — with a correction for the fact that a person's own windows correlate with each other — sharpens the odds-ratio estimate without adding a single subject.

SAMPLING DESIGNS, 7 OF 10

More Controls vs. More Time Windows

In a cohort of 12,090 COPD patients, 245 had a heart attack. Researchers studied its relative risk with antibiotic use in the prior month, using three sampling strategies.

RR 2.00

1 control per case

RR 2.13

10 controls per case

RR 1.99

1 control, 10 windows

95% CIs: 1.16–3.44, 1.48–3.05, 1.36–2.90. Ten monthly windows on the original one control sharpened the estimate about as much as ten times the controls — for far less data-collection cost.

SAMPLING DESIGNS, 8 OF 10

The Case-Cohort Design

First applied by G.B. Hutchison to study leukemia after radiation treatment, and later formalized and named by Ross Prentice, this design also samples from within a cohort, but does not tie each case to its own risk-set.

Step 1

Select all cases of the outcome

Step 2

Randomly draw one fixed-size “subcohort” from the whole cohort, regardless of case status

A patient can be both a case and a member of the random subcohort — that overlap is expected and handled in the analysis.

One Subcohort, Many Uses

Because the random subcohort doesn't depend on any one outcome, it can be reused to study several outcomes at once — something nested case-control cannot do.

The Saskatchewan asthma study needed a control group of 233 for 44 deaths, and a separate group of 422 for 85 near-deaths. Case-cohort could have used one shared subcohort for both. It also lets the time axis switch after the fact, and eases comparison against external population rates.

SAMPLING DESIGNS, 10 OF 10

Nested Case-Control or Case-Cohort?

Nested case-control keeps advantages: simpler sample-size math independent of cohort size, and time-dependent data collected only up to each case's risk-set time, not exhaustively. One benzodiazepine and crash-risk study, designed as case-cohort, had to be redone as nested case-control due to software and hardware limits. It also yields an odds-ratio-style estimate rather than a rate difference — though because it samples from a fully known cohort, absolute rates can sometimes be reconstructed afterward, a “quasi-cohort” approach unavailable to a standalone case-control study.

Rule of thumb: case-cohort suits studies needing multiple outcomes from one sample, or an expensive biomarker in a subgroup. Case-control suits studies needing many exposures, or expensive validation of the outcome itself.

SELF-CONTROLLED DESIGNS, 1 OF 8

Comparing Each Patient to Their Own Earlier Self

Malcolm Maclure's insight: cases are the best representation of the population that produced them. So compare each case to itself at an earlier time. Anything that doesn't change over that time — genetics, chronic conditions, lifelong habits — is automatically controlled for.

This section covers the case-crossover and case-time-control designs, with worked examples, ground rules, and honest limits.

SELF-CONTROLLED DESIGNS, 2 OF 8

The Case-Crossover Design

A case-crossover study is a crossover trial restricted to people who had the event. For each case, researchers check drug use during a short “effect period” right before the event — say, the previous four hours — against that person’s own usual pattern over a longer reference period.

Each case contributes its own exposed-versus-unexposed table for the effect period and the reference period; these combine across all cases, using a Mantel-Haenszel style analysis, into one rate ratio.

SELF-CONTROLLED DESIGNS, 3 OF 8

Worked Example: Beta-Agonists and Heart Rhythm

In a hypothetical study, ten asthma patients had a dangerous fast heart rhythm. Researchers compared beta-agonist use in the four hours before the event against each patient's usual pattern over the past year.

Rate Ratio 3.0

95% CI 1.2–7.6

SELF-CONTROLLED DESIGNS, 4 OF 8

Why the Time Window Matters So Much

In the same heart-rhythm example, changing only the length of the “effect period” before the event swings the result dramatically.

2-hour window

RR 2.0

CI 0.3–12.0

4-hour (main)

RR 3.0

CI 1.2–7.6

6-hour window

RR 5.0

CI 2.0–12.2

All figures are 95% confidence intervals. Getting the effect period right is not a minor detail — it can be the difference between a weak signal and a strong one.

SELF-CONTROLLED DESIGNS, 5 OF 8

Three Rules for Using Case-Crossover Validly

- 1. The outcome must be acute, from a transient drug effect** — not chronic, regular use, and not a slowly-developing side effect.
- 2. The effect period must be precisely determined.** Getting this window even a little wrong changes the answer substantially.
- 3. Reliable data on usual, long-term exposure is needed** — without a trustworthy reference period, there is nothing valid to compare against.

SELF-CONTROLLED DESIGNS, 6 OF 8

Strengths and Limits of Self-Comparison

Strengths

Free of control-selection bias, since patients are their own controls.
Eliminates confounding from anything constant over time.

Limits

Does not automatically control confounders that change over time, and can still reflect confounding by indication if the reason for taking the drug shifts near the event, or case-selection bias if being counted as a case relates to exposure. Imprecise timing also causes information bias; truly non-transient drugs bias results toward no effect.

That last limit showed clearly in a study of long-half-life benzodiazepines and crashes among 5,579 elderly drivers in Quebec: across all cases, OR 0.99 (95% CI 0.83–1.19). Restricted to patients with four or fewer prescriptions in the prior year — truly transient users — OR rose to 1.53 (95% CI 1.08–2.16).

SELF-CONTROLLED DESIGNS, 7 OF 8

The Case-Time-Control Design

Case-crossover assumes exposure prevalence doesn't drift over time — but drug-use patterns often do. This design adds a second, control-subject series, drawn from a conventional case-control study and analyzed the same crossover way.

The within-person, case-only comparison controls stable patient characteristics; the separate control series estimates the secular trend in exposure over time. Combining them — the case-time-control estimate — can reduce both unmeasured confounding by indication and exposure-time trends. It still breaks down if a confounder such as disease severity itself changes over time.

SELF-CONTROLLED DESIGNS, 8 OF 8

Worked Example: Asthma Deaths and Beta-Agonists

Saskatchewan asthma data — 129 fatal or near-fatal cases, 655 matched controls, comparing over 12 vs. 12 or fewer beta-agonist canisters a year.

Method	Odds Ratio (95% CI)
Crude case-control	4.4 (2.9–6.7)
Adjusted for severity	3.1 (1.8–5.4)
Case-crossover	3.2 (1.5–6.8)
Control-crossover	2.6 (1.6–4.1)
Case-time-control	1.2 (0.5–3.0)

The final estimate divides case-crossover by control-crossover, netting out the shared time trend — still assuming no confounder whose effect on treatment choice changes over time.

PROPENSITY SCORES, 1 OF 5

One Number That Stands In for Many Confounders

A propensity score is the estimated probability that a patient would start drug A rather than drug B, given everything recorded before treatment began. It fits the new-user cohort design naturally, and stays efficient even when the outcome is rare.

Scores are usually estimated with logistic regression. Including variables that predict treatment but not outcome only adds noise, and selecting variables by statistical significance is not recommended.

PROPENSITY SCORES, 2 OF 5

Three Ways to Use a Propensity Score

Stratify

Group patients into quintiles or deciles of score, with or without trimming extremes, then compare within groups.

Model

Enter the score directly as a covariate in a regression model of the outcome.

Match

Pair each treated patient with an untreated one of very similar score, then compare pairs.

Matching is the most widely used of the three in pharmacoepidemiology, and is worth a closer look.

PROPENSITY SCORES, 3 OF 5

How Propensity-Score Matching Works

The simplest approach pairs each treated patient with one comparator, though it doesn't use the full dataset. A “caliper” excludes matches beyond a maximum score difference — patients at the extreme edges simply have no good match and are dropped.

Trimming those extremes generally improves internal validity, since unusual scores often reflect atypical, unobserved circumstances. Good matching balances characteristics on average — verified by cross-tabulating them after matching, often summarized with a c-statistic or standardized differences.

PROPSENSITY SCORES, 4 OF 5

When Treatment Choice Becomes Too Predictable

Plotting the two groups' propensity-score distributions side by side is a powerful check. If treatment choice is nearly deterministic given recorded characteristics, the distributions barely overlap — few patients were genuinely comparable candidates for either drug.

Hypothetical illustration: imagine a health plan that covers a fixed combination of ezetimibe and simvastatin only once a patient's cholesterol crosses set thresholds. Those labs would become nearly deterministic predictors of treatment, producing severe separation. As real-world prescribing grows more preference-driven than disease-driven, scores increasingly overlap.

Even a very large study cannot rescue severe separation — the estimate stays imprecise, reflecting a real lack of comparable patients. Strong separation should prompt reconsidering the comparator or population, not accepting an imprecise answer.

PROPENSITY SCORES, 5 OF 5

What Propensity Scores Can and Can't Do

A score can only adjust for measured variables — though absorbing hundreds at once, it can still act as an imperfect proxy for confounders never directly recorded. Using one combined score also gives up seeing each variable's own relationship to the outcome.

More than two treatment options need polytomous models — comparing each to a reference, or matching across several dimensions. When exposure is rare, a “disease risk score,” estimating each patient's outcome probability in the unexposed population, can work better.

CHOOSING COMPARABLE PATIENTS, 1 OF 4

Restriction: Narrowing the Study on Purpose

Restriction narrows the study population so remaining patients are more alike. Some restrictions are explicit — limiting a study to elderly dementia patients, say. Others are implicit, built into a design choice like propensity-score matching.

This section covers three consequential choices: which comparator drug to use, whether to limit to new users, and the resulting trade-off between internal and external validity.

CHOOSING COMPARABLE PATIENTS, 2 OF 4

Picking the Right Comparator Drug

Arguably the single most consequential design decision is which comparator drug to use. The ideal is an identical clinical indication, since indications in secondary data are often unreliable and drugs are frequently used off-label.

Rosiglitazone and pioglitazone, marketed around the same time as similar second-line diabetes drugs, made a natural comparison: treatment choice was nearly random with respect to patient characteristics, producing near-overlapping propensity scores and minimal confounding.

CHOOSING COMPARABLE PATIENTS, 3 OF 4

Restricting to Incident Users of Both Drugs

Limiting a study to new users of both drugs assumes both groups were recently judged by a physician to need treatment. That shared clinical moment makes them more comparable, on both observed and unobserved characteristics.

This is meaningfully fairer than comparing new users to nonusers, or to patients already on treatment for years — comparisons where the two groups may never have faced the same treatment decision at the same disease stage.

CHOOSING COMPARABLE PATIENTS, 4 OF 4

A Trade-Off Researchers Must Own

Propensity-score overlap checks often reveal stretches of the score range with no comparable patients in the opposing group. The usual advice is to trim those patients out, whether by restriction or matching, since they cannot contribute usefully and can introduce their own bias.

Every such restriction improves internal validity — confidence the effect is real — at some cost to generalizability, since the trimmed population is narrower than the drug's real-world use. Pharmacoepidemiology usually prioritizes internal validity, but owes readers transparency about that trade-off.

UNMEASURED CONFOUNDING, 1 OF 7

When a Confounder Was Never Recorded at All

Once a study runs on a fixed dataset, remaining strategies are limited to observable risk factors. Secondary databases often lack exactly the health-status detail that most strongly predicts both treatment choice and outcome, leaving residual confounding.

This section covers proxy adjustment, high-dimensional propensity scores, instrumental variables, and outside data — four ways to account for confounders the main dataset never captured.

UNMEASURED CONFOUNDING, 2 OF 7

Using Proxies to Stand In for What's Missing

Claims and record data can be read as an indirect description of health: old age can stand in for undocumented frailty, home-oxygen dispensing for frailty, regular checkups for health-seeking behavior. A perfect surrogate for a confounder gives full adjustment; a partial one gives partial credit.

Commonly used proxies: number of drugs dispensed, number of physician visits, and hospitalizations before the study start — rough stand-ins for general health, access to care, and monitoring intensity. Studies show these simple proxies meaningfully improve confounding adjustment.

UNMEASURED CONFOUNDING, 3 OF 7

Automating the Search for Proxies

The high-dimensional propensity score algorithm scans a patient's full history of diagnosis, procedure, and drug codes to find hundreds of potential proxy confounders automatically, feeding them into one large model used purely for prediction, not interpretation.

This has repeatedly improved control beyond hand-selected covariates, though it can be less robust with rare outcomes, where score stratification holds up better. A theoretical risk of adjusting for a “pure instrument” exists, but simulations suggest the benefit outweighs it.

UNMEASURED CONFOUNDING, 4 OF 7

Instrumental Variables: Finding Hidden Randomness

Instrumental variable analysis looks for something that pushes treatment choice around in a way that looks random. Classic example: a hospital's formulary. If one stocks only drug A and another only drug B, and patients don't pick hospitals by formulary, drug choice becomes independent of the patient.

1. Strength — the instrument must be associated with which treatment a patient gets.

2. As-good-as-random — the instrument must be unrelated to the patient's own characteristics.

3. Exclusion restriction — it must affect the outcome only through its effect on treatment choice.

The analogy: a trial with imperfect compliance. The instrument is the coin flip; its “strength” is how closely patients stuck to their assigned treatment.

UNMEASURED CONFOUNDING, 5 OF 7

Real Instruments Used in Pharmacoepidemiology

Sudden Shift Over Time

Beta-blockers after a 1998 rule change (Johnston); triamterene after the RALES trial (Juurlink)

Regional Preference

Distance to a heart-catheterization facility (McClellan); regional catheterization rates (Stuckel)

Physician Preference

A doctor's painkiller choice for the prior patient, used to study bleeding risk (Brookhart)

Surgeon Preference

Surgeons who consistently favored aprotinin over aminocaproic acid (Schneeweiss)

Financial Factors

Medication co-payment level and heart-failure drug adherence (Cole)

Institutional Practice

Erythropoietin dosing, nonprofit versus for-profit dialysis centers (Thamer)

UNMEASURED CONFOUNDING, 6 OF 7

The Honest Limits of Instrumental Variables

The surgeon-preference example carries a caveat: it only works if surgeon skill is unrelated to drug preference. That may not hold — academic hospitals could plausibly have looser formularies, sicker patients, and more skilled surgeons all at once.

Tests can flag a clearly weak or biased instrument, but can never confirm one is fully valid — the exclusion restriction is simply not testable. The price of handling unmeasured confounding is an assumption argued on substantive grounds, not proven statistically.

Estimates are also usually less precise, and driven only by the “marginal” patients whose treatment the instrument actually swayed — much like a propensity-score analysis is informed only by patients in the region of good overlap.

UNMEASURED CONFOUNDING, 7 OF 7

Bringing in Outside Data to Fill the Gaps

Two-Stage Sampling

Sample patients by both exposure and outcome status, collect richer confounder data only in that subsample, then reweight the results back to represent the full study population.

Go Collect It Yourself

A nested case-control sample can be purpose-built for richer data on a subset, or linked to registries to check whether claims-invisible traits — labs, BMI — were balanced by claims-only matching. Tailored, but slow and labor-intensive.

Borrow Existing Data

A survey like the Medicare Current Beneficiary Survey, capturing impairment not in claims, can adjust an analysis faster if representative. Propensity score calibration extends this to several confounders using an “error-prone” and a “complete” score.

STRESS-TESTING THE FINDINGS, 1 OF 4

How Strong Would a Hidden Confounder Need to Be?

A valuable, underused check: how strong and how imbalanced would an unmeasured confounder need to be to fully explain away an observed result? Researchers can calculate this directly, working backward from the effect observed.

Free spreadsheet tools exist for this. Timothy Lash and Aliza Fink proposed combining it with sensitivity analyses for misclassification and selection bias, into one unified framework.

STRESS-TESTING THE FINDINGS, 2 OF 4

A Real Example: A Result That Held Up

A cohort study of TNF-alpha inhibitors for rheumatoid arthritis found no expected link to serious bacterial infections — raising a worry that physicians were simply steering the drug away from sicker patients.

The sensitivity calculation found that a hidden confounder would need to be both unusually strong and unusually imbalanced to erase the null result — judged unlikely. The finding stood: sometimes a sensitivity analysis simply confirms a result is more robust than it first appears.

STRESS-TESTING THE FINDINGS, 3 OF 4

How Long a Washout Period Is Long Enough?

Databases can't ask when a patient first started a drug, so "new users" are found indirectly: no prior dispensing during a look-back window, typically six months. Testing that window — extending to nine or twelve months — is worthwhile.

A pediatric antidepressant safety study in British Columbia extended its washout from one year to three specifically to ensure treatment-naivety, which helped balance the groups. Longer washouts trade off against sample size. The exposure "risk window" carries the same uncertainty and deserves the same testing.

STRESS-TESTING THE FINDINGS, 4 OF 4

Two Ways to Handle Patients Who Stop Treatment

As-Treated

Stop counting follow-up when a patient stops or switches. Biased if the reason for stopping — lack of effect, early side effects — relates to the outcome.

Cumulative Risk

Keep the original exposure group regardless of later changes, like intention-to-treat. Avoids that bias, but adds exposure misclassification, usually toward the null.

Running both together brackets a plausible range for the true effect. Properly adjusting for nonadherence needs data on why patients stopped — often unavailable with enough accuracy.

CONCLUDING REMARKS, 1 OF 3

Three Directions Worth Watching

Minimizing confounding in nonrandomized pharmacoepidemiologic research is an ongoing, unfinished project — much work remains before these methods can reliably carry the weight of real medical decisions.

Instrumental variables that exploit the layered structure of healthcare systems — patient, provider, region.

Propensity-score methods paired with data-mining, to find high-dimensional proxy confounders automatically.

Enriching claims data with clinical detail from records, registries, surveys, and lab repositories.

CONCLUDING REMARKS, 2 OF 3

Richer Data Brings Its Own New Problem

The most promising direction — enriching routine claims data with real clinical detail — comes with an equally large catch.

Information from routine care is often itself selectively recorded — ordered for the patients a clinician already suspects need attention. Richer data can quietly introduce its own new selection and confounding problems even as it solves the old ones.

CONCLUDING REMARKS, 3 OF 3

No Single Method Solves Confounding Alone

From how a cohort is assembled, to how patients are sampled, to how a propensity score is built, to which comparator is chosen, to the instrumental variables and sensitivity checks brought in when data runs short — every tool here tackles the same problem from a different angle. Used together, and honestly, they get pharmacoepidemiology as close as it can to knowing what would have happened to the same patient, treated differently.

Adapted for a general audience by Omokhodion Alfred Eriakha, from “Advanced Approaches to Controlling Confounding in Pharmacoepidemiologic Studies” by Sebastian Schneeweiss and Samy Suissa, in Pharmacoepidemiology, 6th Edition.